

<u>Step</u>	<u>Reason</u>
1. $\exists x(S(x) \wedge \neg B(x))$	Premise
2. $\forall x(S(x) \Rightarrow E(x))$	Premise
3. $S(c) \wedge \neg B(c)$	Existential instantiation on 1
4. $S(c)$	Simplification on 3
5. $\neg B(c)$	Simplification on 3
6. $S(c) \Rightarrow E(c)$	Universal instantiation on 2
7. $E(c)$	Modus ponens on 4, 6
8. $E(c) \wedge \neg B(c)$	Conjunction on 7, 5
9. $\exists x(E(x) \wedge \neg B(x))$	Existential generalization on 8

1.6.9 Combining Rules of Inference for Propositions and Quantified Statements

Universal modus ponens: This rule tells us that if $\forall x(P(x) \rightarrow Q(x))$ is true, and if $P(a)$ is true for a particular element a in the domain of the universal quantifier, then $Q(a)$ must also be true.

To see this, note that by universal instantiation, $P(a) \rightarrow Q(a)$ is True. Then, by modus ponens, $Q(a)$ must also be True.

We can describe universal modus ponens as follows:

$\forall x(P(x) \rightarrow Q(x))$
 $P(a)$, where a is a particular element in the domain
 $\therefore Q(a)$

Universal modus tollens: Universal modus tollens combines universal instantiation and modus tollens and can be expressed in the following way:

$\forall x(P(x) \rightarrow Q(x))$
 $\neg Q(a)$, where a is a particular element in the domain
 $\therefore \neg P(a)$